

1 RESULTS – four queued tests against ZL3b-n (full corpus, 32,184 tokens, v4 tokenizer)

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4 TEST 3 – Herbal-A entry-header template (gallows-initial paragraph 1st token)

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7	Section	#paras	para-open	gallows%	baseline	gallows%	ratio
8	herbal-A	118	74.6%		9.7%		7.70x
9	herbal-B	28	96.4%		10.6%		9.13x
10	q13	96	24.0%		5.6%		4.27x
11	q20	28	96.4%		8.1%		11.86x
12	pharm	150	10.7%		5.3%		2.03x
13	zodiac	22	0.0%		3.9%		0.00x
14	cosmo	71	1.4%		5.6%		0.25x

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16 VERDICT – STRONG. Paragraph-opening slot is a real, decorated, section-specific
17 convention. Herbal-A 7.7x baseline. Q20 11.9x (also strongly templated).
18 Pharm only 2.0x – pharmacy paragraphs do NOT use the same entry-header convention.
19 Q13 4.3x – mild template, less ceremonial than herbal-A.
20 Zodiac & cosmo – zero or near-zero (these aren't paragraph-of-encyclopedia objects;
21 they're labels and panel-text).

22

23 NEW finding: Herbal-B (96.4%) and Q20 (96.4%) hit ceiling. Stronger templates than
24 herbal-A. Q20 was previously framed as "stars/recipes" – its near-universal
25 gallows-initial paragraphs argue for it being a SECOND encyclopedic entry-format,
26 not just narrative text.

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28 Pharmacy genre confirmed DIFFERENT. Per-paragraph entry-header is herbal-tradition,
29 not pharma-tradition. Pharmacy paragraphs open with non-gallows morphology – they
30 look like recipe procedures, not lemma entries. (This is the three-genre answer:
31 herbal-A = de simplicibus, Q20 = a second decorated-entry genre, pharm = procedures.)

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34 TEST 4 – Q13 vs Q20 register comparison

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37		q13	q20	pharm	herb-A	herb-B	cosmo
38	n	6377	11602	1293	7762	3116	478
39	chedy/k	28.9	17.1	0.0	0.0	18.9	0.0
40	shedy/k	34.3	10.1	0.8	0.0	10.3	0.0
41	qok-/k	159.3	100.5	68.8	31.6	62.6	6.3
42	chor/k	1.3	4.1	23.2	46.1	3.2	8.4
43	chol/k	3.9	12.2	41.8	54.9	6.4	14.6
44	ot-/k	45.2	72.6	44.9	42.4	48.1	77.4
45	ok-/k	49.4	69.6	93.6	37.0	82.8	106.7

46	suf -edy	25.7%	13.7%	0.4%	0.1%	14.0%	0.4%
47	suf -dy	30.0%	18.8%	8.5%	6.0%	26.8%	8.8%

48

49 VERDICT – Q13 and Q20 are RELATED but NOT identical registers.

50 - Both share: high qok-, high -edy/-dy suffix, low chor.

51 - They differ: chedy/shedy/qok are TWICE as dense in Q13.

52 - Q20 is INTERMEDIATE between Q13 and pharmacy/herbal-A.

53 - Q20 has more ot-/ok- (substance prefixes) than Q13 – more substance content.

54

55 Reading: Q20 = process-prescribed-with-substances (recipe-procedure-with-ingredients).

56 Q13 = pure-process (balneology / humoral-process language).

57 Pharm = substance-procedure (chor/chol-rich, low chedy, recipe).

58

59 This refines the May 11 LOCKED finding. "Process register" is a continuum:

60 Q13 is process-heavy pole, Q20 is process+substance, pharm is substance-heavy pole.

61

62 The herbal-B intermediate result (chedy 18.9, chor 3.2, -dy 26.8%) shows herbal-B

63 is ALSO in this continuum – much closer to Q13 than herbal-A is.

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65 =====

66 TEST 1 – Adjacency: dar, dor, ar, or, y-, s, r – clitic vs free function word

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69 Key question: does dar/dor heavily favor one stem class as its right neighbor?

70 (Clitic = yes, attached to specific class. Free function = no, distributes evenly.)

71

72 dar (n=253)

73 following: ch/sh 24.5%, other 23.7%, short 14.2%, ot/ok 9.1%, ched/shed 5.5%

74 preceding: other 22.9%, qok/qot 17.0%, ch/sh 12.6%, ot/ok 12.6%

75 -> dar prefers to be FOLLOWED by ch/sh stems (24.5% vs ~10% baseline for ch/sh).

76 Mild but real bias. NOT a tight clitic – distributes across many classes.

77

78 dor (n=61)

79 following: ch/sh 37.7%, other 23.0%, short 14.8%, chor/chol 9.8%

80 -> dor STRONGLY prefers ch/sh stems following (37.7%).

81 Also notable: chor/chol 9.8% (5x dar's 2.0%). dor connects to substance words.

82

83 ar (n=302)

84 following: other 37.4%, short 23.2%, ch/sh 12.9%, ot/ok 11.6%

85 -> ar distributes widely; following is mostly "other" (uncategorized rare types).

86 Looks like a SUFFIX/CLITIC that has been DETACHED from another word

87 (an enclitic that sometimes appears as a free token).

88

89 or (n=300)

90 following: short 30.0%, other 27.7%, ch/sh 23.0%

-> or strongly prefers SHORT/CLOSED-class following (30.0%).

This is the signature of a COORDINATOR – sits between phrases, next token is often another function word starting a new phrase.

ol (n=464)

following: ch/sh 25.0%, other 22.2%, short 12.1%, ched/shed 11.4%, t/k 10.8%

preceding: other 29.7%, ch/sh 15.1%, short 13.4%, qok/qot 12.3%, ot/ok 11.9%

-> ol is the most evenly-distributed of all candidates – almost neutral.

Behaves like a TRUE free function word (prepositional / general particle).

y (n=263)

following: t/k 22.1%, ch/sh 22.1%, other 16.0%

preceding: other 24.3%, short 17.1%

-> y- strongly prefers GALLOWS-INITIAL (t/k) AND ch/sh following.

This is the proclitic signature. y- attaches to the front of words.

Word-initial y- (ydain, ychor, ytain) is the same morpheme detached.

STRONG candidate for proclitic prefix (negation, topicalizer, or definite article).

s (n=277)

following: short 35.4%, other 22.7%, ch/sh 17.7%

-> s prefers short tokens following – looks like a clause/phrase boundary

marker. NOT a content morpheme.

r (n=148)

following: short 38.5%, ch/sh 25.7%, other 20.3%

-> r similar to s. Boundary/punctuation-like behavior.

VERDICT on the clitic hypothesis:

dar – STRONG CANDIDATE for prepositional clitic (favors ch/sh stems following).

dor – STRONGER candidate (37.7% ch/sh, also 9.8% chor/chol). Closer to a true adposition + content noun pattern.

ar – looks like an enclitic that surfaces as free token in some environments (the high "other" + "short" rate means it appears in fragment-like positions).

or – coordinator signature (sits between phrases, next-token = function word).

ol – neutral / true free particle.

y- – proclitic prefix attaching to gallows-initial and ch/sh words.

s/r – boundary markers / phrase-terminators.

The "Semitic-style prepositional clitic" hypothesis SURVIVES, specifically for dar and dor. ar/or/ol behave differently – they are coordinator-like and particle-like. The system is not monolithic – different short morphemes do different grammatical work.

TEST 2 – a/o vowel-grade contrariety (parallel to o/e)

92 a/o minimal pairs both occurring >=5 times. Mean a-bias: -0.019 (essentially zero).
40 a-dominant, 48 o-dominant. No global directional flip.

But INDIVIDUAL pairs are sharply polarized:

daiin (721) vs doiin (11)	a-bias +0.97	- STRUCTURAL ASYMMETRY
qokaiin (242) vs qokoiin (6)	a-bias +0.95	- same
okaiin (193) vs okoiin (6)	a-bias +0.94	- same
aiin (396) vs oiin (27)	a-bias +0.87	- same
ain (97) vs oin (7)	a-bias +0.87	- same
vs.		
chol (346) vs chal (44)	a-bias -0.77	
shol (167) vs shal (15)	a-bias -0.84	
o (157) vs a (9)	a-bias -0.89	

So the system is: a-grade DOMINATES in -ain/-aiin suffixed forms.

o-grade DOMINATES in -ol/-ar bare-or-vowel-ending substance words.

This is NOT the same as o↔e contrariety (which DID show a clean directional flip
9/9 same direction). The a/o split is a structural co-occurrence pattern, not
a contrastive ablaut rule.

VERDICT: a/o vowel-grade contrariety NOT supported as a productive paired
contrariety. CONFIRMED FALSIFIED. The May 11 locked o↔e rule remains the
only operative vowel-grade contrariety. The a in -ain/-aiin and the o in
-ol/-or are sitting in different morphological slots, not paired alternants.

SUMMARY – what was locked, what changed, what's new

LOCKED (new, this session):

- Herbal-A 74.6% gallows paragraph-opening (7.7x baseline)
- Q20 96.4% gallows paragraph-opening (11.9x baseline) – newly-strong template
- Herbal-B 96.4% gallows paragraph-opening (9.1x baseline)
- Pharm DOES NOT use the herbal entry-header convention (10.7%, only 2.0x baseline)
- dor preferentially followed by ch/sh stems (37.7%) – clitic-like
- y- proclitic prefix to gallows + ch/sh stems (22% + 22% = 44%)

REFINED:

- "Process register" is a continuum, not binary. Q13 = process pole.
Pharm = substance pole. Q20 = process+substance intermediate.
Herbal-B = intermediate-process. Herbal-A = substance/de-simplicibus.

181 FALSIFIED:
182 - a/o as productive paired contrariety (confirmed, no clean flip)
183 - eey/eedy as degree particles (confirmed predicate suffix)
184
185 NEW HYPOTHESES (CANDIDATE):
186 - Three encyclopedic-text genres confirmed by template:
187 (1) herbal-A / herbal-B / Q20 = entry-header genre (decorated initial)
188 (2) Q13 = continuous-process genre (low decoration, predicate-dense)
189 (3) pharm = recipe-procedure genre (low decoration, substance-dense)
190 - dar / dor are prepositional clitics in the Semitic style. Distinct functions
191 likely: dor = with substance / from substance (ch/sh + chor/chol affinity).
192 dar = more general adposition. TESTABLE.
193